Explicit coset examples and their full triality centralizers

Human Compression project: recovery of L122

The finite examples of Leemans and Stokes [1] are genuine mathematical claims, including their full automorphism groups. The original does not give its subgroup markings or software. We supply complete explicit witnesses for every listed parameter. We do not identify these witnesses with the unprovided historical choices. The complete companion *Coset trialities, chamber recognition and reduced relations* supplies the construction, flag-completion and full-automorphism arguments used below.

1 Exact finite-field markings

For each field below, z is the residue of X in the indicated polynomial quotient. These are the primitive Conway generators used by GAP; the polynomial coefficients also specify the fields independently of that name.

q	defining polynomial
8	$X^3 + X + 1$
27	$X^3 + 2X + 1$
64	$X^6 + X^4 + X^3 + X + 1$
125	$X^3 + 3X + 3$
512	$X^9 + X^4 + 1$

For $q = 8, 27, 64, 125, 512$, use the natural projective action of $\mathrm{SL}_2(q)$ on the $q+1$ lines represented by $(1, x)$ and $(0, 1)$. Its image is $G = \mathrm{PSL}_2(q)$. The following matrices of determinant one specify the first projective involution r_0:

q	R_0
8	$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$
27	$\begin{pmatrix} z^2 & 1 \\ z^5 & z^{15} \end{pmatrix}$
64	$\begin{pmatrix} z^{60} & z^{11} \\ z^9 & z^{60} \end{pmatrix}$
125	$\begin{pmatrix} z & 1 \\ z^9 & z^{63} \end{pmatrix}$
512	$\begin{pmatrix} z^{507} & z^9 \\ z & z^{507} \end{pmatrix}$

For $q = 27, 64, 125, 512$, put respectively $p = 3, 4, 5, 8$ and let σ be the field automorphism raising every entry to its pth power. Set $r_i = \sigma^i(r_0)$ for $i = 0, 1, 2$. For $q = 8$ use instead conjugation by the projectivity represented by

$$T = \begin{pmatrix} z^5 & z^5 \\ z & z^4 \end{pmatrix},$$

which has projective order three. This preserves the source's explicit inner/outer distinction. Set $H_i = \langle r_j, r_k \rangle$ for the complementary pair.

For the Suzuki example use the standard four-dimensional matrix group Sz(8), acting on its 65-point ovoid, with the same generator $z \in \mathbb{F}_8$. Here take

$$R_0 = \begin{pmatrix} z^6 & z^3 & z^4 & 1 \\ z^6 & z^6 & z^3 & z^4 \\ z^5 & z & z^6 & z^3 \\ z^4 & z^5 & z^6 & z^6 \end{pmatrix}$$

and let σ square every entry. Again use its three cyclic images and the complementary pair subgroups. Standard Suzuki coordinates and the faithful ovoid identification are retained classical inputs, already used in the earlier Suzuki work; the exact standard matrices and ovoid are those of Bray and Bäärnhielm [4], equations (1)–(5); no group is identified merely by its order.

2 The whole finite certificate and geometric conclusions

The required file `programs/verify_examples.g`, together with the whole marking data `data/seeds.g`, checks the six cases. For the projective linear groups it constructs every projective point and its matrix action. For the Suzuki group it constructs the full 65-point orbit of $(1, 0, 0, 0)$. In every case the permutation order is verified on a fresh group formed solely from copies of the permutation generators, with no cached size, chain, ambient parent or supplied order bound. Stabilizer-chain verification uses `random=1000`; internal random construction is not excluded. The matrix group models, exact finite-field and subgroup operations are explicit GAP trust dependencies.

The program checks every seed's membership and involution order, full generation, order-three automorphism, cyclic transport of all three parabolics, all 64 subset intersections, and each of the three product intersections $H_i \cap H_j H_k$. The latter have exactly four elements. It obtains the following full results; $m = |r_0 r_1|$, and v is the number of objects of each type.

G	$\lvert G\rvert$	$\lvert H_i\rvert$	m	v	$\lvert G^\sigma\rvert$
Sz(8)	29120	14	7	2080	20
$\mathrm{PSL}_2(8)$	504	14	7	36	9
$\mathrm{PSL}_2(27)$	9828	28	14	351	12
$\mathrm{PSL}_2(64)$	262080	130	65	2016	60
$\mathrm{PSL}_2(125)$	976500	62	31	15750	60
$\mathrm{PSL}_2(512)$	134217216	1026	513	130816	504

These values and the verified equalities are computational proof premises; there is no reliance on an unprovided historical Magma run. The subsequent passage to every flag and every automorphism is written, not sampled.

Theorem 1. *For all six markings the coset system is a thin, residually connected, flag-transitive rank-three geometry with a triality induced by σ. Its full type-preserving automorphism group is G. Its full reduced oriented graph automorphism group is G^σ.*

Proof. The certificates give exactly the hypotheses of the companion's chamber recognition theorem, including trivial chamber stabilizer and the complete four-element product identity. That theorem proves all the geometric properties and that no additional typed automorphism exists. The cyclic parabolic transport and $\sigma^3 = 1$ give an actual triality. The companion's reconstruction and centralizer theorem then proves the last assertion. No inference from the order of a discovered automorphism subgroup to the order of the full graph group is made. □

The outer qualification is also checked. On each natural point set the program constructs the field permutation, verifies its conjugation on all group generators, checks that it is outside G,

and verifies that the point stabilizer is transitive on the remaining points. A doubly transitive permutation group on at least three points has trivial centralizer: a centralizing permutation sending x to a different y would force the point stabilizer at x to fix y, contrary to that transitivity. A centralizing permutation fixing x fixes every point. Thus a field permutation outside G cannot induce an inner automorphism. This proves the four projective-linear and the Suzuki outer assertions.

For $q = p^3$ in the four outer projective linear cases, the fixed group is precisely $\mathrm{PSL}_2(p)$. Indeed if a projectivity represented by $A \in \mathrm{SL}_2(q)$ is fixed, then $A^{(p)} = \epsilon A$, where $\epsilon = \pm 1$ in odd characteristic and $\epsilon = 1$ in characteristic two. Three iterations give $\epsilon^3 = 1$, so in both cases $\epsilon = 1$. Thus every entry belongs to $\mathbb{F}_p$, and the converse is immediate. The scalar-center quotient gives exactly the asserted group. The program also constructs the embedded $\mathrm{SL}_2(p)$ image, verifies its order and checks that each of its generators is fixed. For $q = 8$ the centralizer of the specified inner element is cyclic of order nine; the program checks its actual subgroup and cyclicity. This explains the source's exception to the subfield pattern.

For the Suzuki case, the field-fixed group is $\mathrm{Sz}(2)$ of order 20. This can be seen from the unique standard Bruhat parameters: matrices fixed by squaring have parameters in $\mathbb{F}_2$, giving $2^2+2^4 = 20$ elements. This is the same retained matrix/Bruhat identification used in the earlier Suzuki field-intersection proof. The certificate independently checks all 29120 permutation elements, finding exactly 20 fixed ones. It also constructs the binary-coordinate subgroup, checks its order 20, its normal cyclic subgroup of order 5, and a cyclic complement of order 4 acting faithfully. It is therefore $C_5 : C_4$, as required, rather than a dihedral group of order 20.

3 The absolute subgeometry in the Suzuki case

An object $H_0 g$ is absolute precisely when

$$g\sigma(g)^{-1} \in H_0 H_1.$$

This is just the right-coset intersection criterion for $H_0 g$ and $H_1 \sigma(g)$. The program tests every one of the 2080 cosets and obtains exactly 20 absolute objects. The order 20 fixed subgroup has one orbit on them, necessarily regular. By reconstruction each produces a reduced loop and a triple of pairwise incident objects in the original geometry. The induced system on all 60 of these objects is checked in full: it has 60 edges and 20 chambers, and every edge extends. More precisely it is a disjoint union of 20 chambers. Thus it is a subgeometry with the stated fixed-subgroup action; connectedness of this subgeometry was not claimed by the source and is not silently added. The full incidence graph has 6240 vertices and the reduced oriented graph 2080 vertices, as in [1].

4 The Mathieu example and why the full group matters

Retain the classical degree-twelve, three-transitive action of M_{11} on the cosets of a $\mathrm{PSL}_2(11)$ subgroup, and the classical fact $\mathrm{Aut}(M_{11}) = M_{11}$. These named-group facts are separate from the following elementary incidence calculation.

Proposition 2. *Let $G = M_{11}$ in that action on Ω, $|\Omega| = 12$, and take H_i to stabilize three distinct points. The resulting geometry consists of three typed copies of Ω, with distinct labels incident. It is firm, residually connected and flag-transitive, has full typed group S_{12} and full type-permuting group $S_{12} \times S_3$, and admits a triality induced by an inner automorphism of G.*

Proof. Identify a coset of a point stabilizer with the image of its point. Two cosets meet exactly when one element of G realizes both prescribed images. Two-transitivity says this happens exactly for distinct target labels. Three-transitivity gives transitivity on chambers, which are precisely triples of distinct labels. Every flag extends and every panel has ten completions, proving firmness. A vertex residue is the bipartite graph on two copies of the other eleven labels, with all unequal

pairs joined. This graph is connected: any two vertices in the same part have a common neighbor, and the opposite-part case follows by a path of length at most three. The full graph is connected as well. These are all residues of rank at least two.

A typed automorphism is a triple of permutations of Ω. Nonincidence between two different types is exactly equality of labels, so the three permutations must coincide. Conversely every simultaneous permutation preserves incidence. Arbitrary permutations of the three types also preserve incidence and commute with the simultaneous label permutations. This proves the two full group assertions, including the absence of any additional correlations.

There is an element of order three in M_{11}, and its faithful action has a three-cycle. By three-transitivity, conjugate that cycle to the three chosen points. Conjugation by this element cyclically permutes their stabilizers and has order three. The coset construction therefore gives the required inner triality. The larger S_{12} typed group shows why one cannot replace the full-group proof of Theorem 1 by merely exhibiting the original coset action. □

5 Source contract and remaining qualification

All three finite example sections of [1] are now supplied with explicit models and proofs of their full-group claims. The historical software mentioned there is Magma, Sage and GAP; no source executable listing is supplied. Our complete required certificate and data are separate from any future manuscript-length comparison. No software cost is declared zero, and no factor or full formalization is claimed.

The companion corrects the later higher-order normalization assertions with explicit counterexamples and gives the lossless order-r voltage convention. The stronger literal reading of Lemma 1 without an order condition remains separate: cycling parabolics proves an induced correlation with the specified type cycle, but the argument alone does not show order three. Our examples satisfy the necessary order condition. No complete-source admission is claimed while the general bare existential reading is unresolved. The source's proposed future exploration of other r-alities is not treated as a solved question, and no historical novelty is inferred from these repairs.

References

[1] D. Leemans and K. Stokes, *Coset geometries with trialities and their reduced incidence graphs*, Acta Math. Univ. Comenianae **88**(3) (2019), 911–916.

[2] M. Suzuki, *A new type of simple groups of finite order*, Proc. Natl. Acad. Sci. USA **46** (1960), 868–870.

[3] R. A. Wilson, *On the simple groups of Suzuki and Ree*, Proc. Lond. Math. Soc. **107** (2013), 680–712.

[4] J. N. Bray and H. Bäärnhielm, *A new method for recognising Suzuki groups*, arXiv:1706.05697v1 (2017), equations (1)–(5). Auxiliary classical model, not an additional original.

[5] The GAP Group, *GAP*, version 4.15.1. Required exact permutation, finite-field and subgroup operations; named model identities and full runtime costs remain explicit external inputs.